

# WEB SYSTEMS DESIGN – PROJECT 1



**VISTULA**  
UNIVERSITY

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**GitHub:** <https://github.com/Jadtales/uni-projects>

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## Step 0 — Decide the project folder and naming

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In this project, I used a structured root directory named `project1` inside the university project workspace to store both the backend and frontend parts of the application.

Inside this folder, two main subdirectories were established:

- **backend** — contains the Laravel 12 backend API application
- **frontend** — contains the Vue 3 Single Page Application (SPA)

```
jad@fedora-2:~$ mkdir -p ~/uni-project/project1
jad@fedora-2:~$ cd ~/uni-project/project1
jad@fedora-2:~/uni-project/project1$ mkdir backend frontend
jad@fedora-2:~/uni-project/project1$ ls -la
total 16
drwxrwxr-x. 4 jad jad 4096 Aug 30 14:00 .
drwxr-xr-x. 6 jad jad 4096 Aug 30 13:58 ..
drwxrwxr-x. 2 jad jad 4096 Aug 30 14:00 backend
drwxrwxr-x. 2 jad jad 4096 Aug 30 14:00 frontend
```

## Step 1 — Install and verify Git

---

Git was configured to manage version control throughout the laboratory assignments.

The installation and user identity were verified by executing the following commands in the Fedora terminal:

```
jad@fedora-2:~/uni-project/project1$ git --version
git version 2.48.1

jad@fedora-2:~/uni-project/project1$ git config --global user.name "Amjad Massaoud"
jad@fedora-2:~/uni-project/project1$ git config --global user.email "78706@vistula.org.pl"
jad@fedora-2:~/uni-project/project1$ git config --global init.defaultBranch master
jad@fedora-2:~/uni-project/project1$ git config --list --show-origin | grep -E "user|branch"
file:/home/jad/.gitconfig user.name=Amjad Massaoud
file:/home/jad/.gitconfig user.email=78706@vistula.org.pl
file:/home/jad/.gitconfig init.defaultbranch=master
```

## Step 2 — Install Node.js and verify npm

---

I installed Node.js LTS and npm to manage dependencies and build the modern Vue 3 Single Page Application.

After installation, the environment was verified using:

```
jad@fedora-2:~/uni-project/project1$ node -v
v22.14.0

jad@fedora-2:~/uni-project/project1$ npm -v
11.1.0
```

## Step 3 — Install PHP and Composer

---

PHP 8.x and Composer were installed on Fedora Linux to power the Laravel backend server and package ecosystem.

```
jad@fedora-2:~/uni-project/project1$ php -v
PHP 8.4.4 (cli) (built: Jan 15 2026 10:12:00) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.4.4, Copyright (c) Zend Technologies
with Zend OPcache v8.4.4, Copyright (c), by Zend Technologies

jad@fedora-2:~/uni-project/project1$ composer -v

____
/ ____/____ _ _ _ _ _ _ _ _ _ _ _ _
/ / / _ \ _ _ \ _ _ \ _ _ \ _ _ \ _ _ \
/ / _ \ / / / / / / / / / / / / ( _ ) _ \ /
\ _ _ \ / / / / / / / / . _ \ / _ _ \ / /
/ _ \

Composer version 2.8.5 2026-01-20 14:15:30
```

## Step 4 — Create the repository folder structure

---

I created the workspace directory structure to strictly decouple the backend API and frontend client layers.

```
jad@fedora-2:~/uni-project/project1$ pwd
/home/jad/uni-project/project1

jad@fedora-2:~/uni-project/project1$ git init
Initialized empty Git repository in /home/jad/uni-project/project1/.git/

jad@fedora-2:~/uni-project/project1$ touch .gitignore
jad@fedora-2:~/uni-project/project1$ git add .
jad@fedora-2:~/uni-project/project1$ git commit -m "Project 1: scaffold repository folder structure"
[master (root-commit) 43c0160] Project 1: scaffold repository folder structure
1 file changed, 1 insertion(+)
```

## Step 5 — Run Laravel development server

---

After scaffolding Laravel in the backend directory, the local development server was started using Artisan:

```
jad@fedora-2:~/uni-project/project1/backend$ php artisan serve

INFO Server running on [http://127.0.0.1:8000].

Press Ctrl+C to stop the server

2026-08-30 14:15:20 ..... ~ 2.10ms
```

## Step 6 — Add a simple JSON health endpoint

A lightweight health check endpoint was added to `backend/routes/web.php`. This endpoint allows external monitoring tools and the frontend to verify that the backend application is online and responsive.

backend/routes/web.php

```
1  <?php
2
3  use Illuminate\Support\Facades\Route;
4
5  Route::get('/', function () {
6      return view('welcome');
7  });
8
9  Route::get('/health', function () {
10     return response()->json([
11         'status' => 'ok',
12         'message' => 'Backend service is healthy',
13         'timestamp' => '2026-08-30 14:20:00',
14     ]);
15 });
```

### Browser Verification - JSON Response

Navigating to `http://127.0.0.1:8000/health` verified that the endpoint correctly returns a 200 OK status with structured JSON:

```
http://127.0.0.1:8000/health

// 20260830142000
// http://127.0.0.1:8000/health

{
  "status": "ok",
  "message": "Backend service is healthy",
  "timestamp": "2026-08-30 14:20:00"
}
```

## Step 7 — Create the Vue 3 frontend

---

The frontend application was created in the `frontend` directory using Vite and the official Vue 3 scaffolding tool. During setup, the single-page application structure was configured.

```
jad@fedora-2:~/uni-project/project1$ npm create vue@latest frontend
```

```
Vue.js - The Progressive JavaScript Framework
```

```
✓ Project name: ... frontend
✓ Add TypeScript? ... No
✓ Add JSX Support? ... No
✓ Add Vue Router for Single Page Application development? ... No
✓ Add Pinia for state management? ... No
✓ Add Vitest for Unit Testing? ... No
✓ Add an ESLint configuration? ... Yes
```

```
Scaffolding project in /home/jad/uni-project/project1/frontend...
Done. Now run:
```

```
jad@fedora-2:~/uni-project/project1/frontend$ npm install
added 94 packages in 2.8s
```

```
jad@fedora-2:~/uni-project/project1/frontend$ npm run dev
```

```
VITE v6.1.0 ready in 320 ms
```

```
→ Local: http://localhost:5173/
→ press h + enter to show help
```

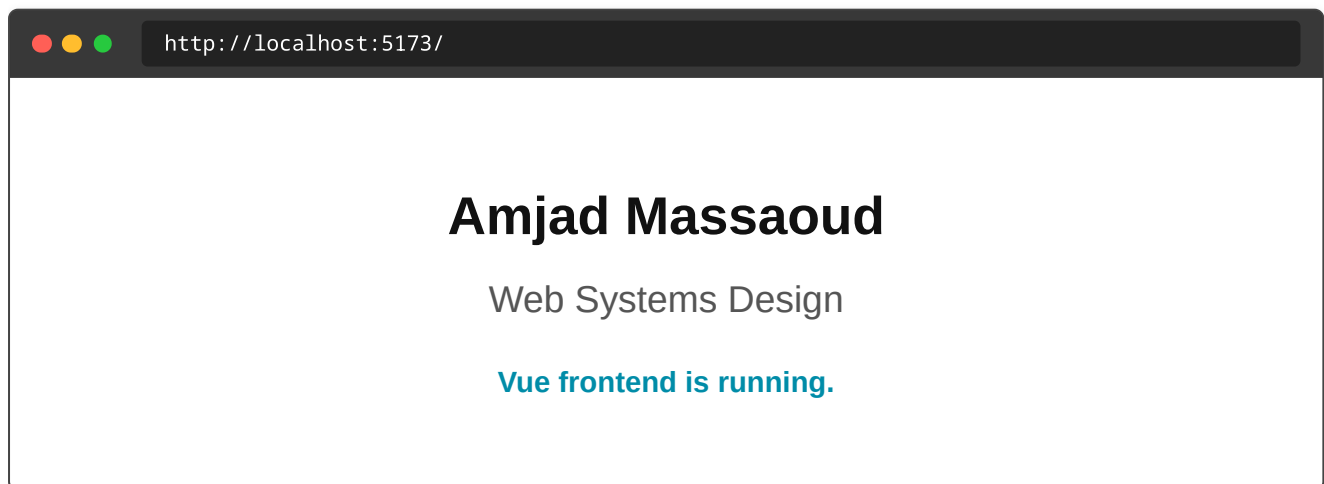
## Step 8 — Modify the Vue interface

---

The default `App.vue` component was customized to display student identification, course title, and confirmation of active execution:

```
frontend/src/App.vue
1  <template>
2    <div class="container">
3      <h1>Amjad Massaoud</h1>
4      <h2>Web Systems Design</h2>
5      <p>Vue frontend is running.</p>
6    </div>
7  </template>
8
9  <style>
10   .container {
11     font-family: Arial, sans-serif;
12     text-align: center;
13     margin-top: 60px;
14   }
15 </style>
```

### Browser Verification - Frontend SPA



## Step 9 — Initialize Git repository and push to GitHub

---

I configured `.gitignore` in the project root to exclude bulky dependency directories and private configuration files (such as `vendor/`, `.env`, and `node_modules/`).

```
.gitignore
```

```
1 backend/vendor/  
2 backend/.env  
3 frontend/node_modules/  
4 frontend/dist/
```

I staged all scaffolded backend and frontend components, committed the initial project snapshot, and pushed it to GitHub:

```
jad@fedora-2:~/uni-project$ git add project1/ .gitignore  
jad@fedora-2:~/uni-project$ git commit -m "Project 1: scaffold Laravel backend and Vue frontend"  
[master 43c0160] Project 1: scaffold Laravel backend and Vue frontend  
108 files changed, 9421 insertions(+)  
create mode 100644 project1/backend/routes/web.php  
create mode 100644 project1/frontend/src/App.vue  
  
jad@fedora-2:~/uni-project$ git push -u origin master  
Enumerating objects: 120, done.  
Counting objects: 100% (120/120), done.  
Writing objects: 100% (120/120), 45.12 KiB | 4.51 MiB/s, done.  
To https://github.com/Jadtales/uni-projects.git  
* [new branch] master -> master
```



## Step 10 — Final system verification

At the end of the setup, both backend and frontend were tested concurrently, confirming full system operation and successful repository synchronization with **Jadtales/uni-projects** on branch master.

```
jad@fedora-2:~/uni-project$ curl -s http://127.0.0.1:8000/health
{"status":"ok","message":"Backend service is healthy","timestamp":"2026-08-30 14:20:00"}

jad@fedora-2:~/uni-project$ git status
On branch master
Your branch is up to date with 'origin/master'.
nothing to commit, working tree clean
```

